CSA12 - Computer Architecture

Question 3: Processor Benchmarking Tool

Problem Statement

Develop a benchmarking tool that accepts Instruction Count (IC), Clock Cycle Time, Clock Frequency and CPI for multiple processors. Compute CPU Execution Time, MIPS, IPC and Speedup, then compare processor performance.

Objective

To evaluate processor performance using standard benchmarking parameters and compare multiple processors based on execution time and efficiency.

Theory

Processor benchmarking measures CPU performance using metrics such as: Instruction Count (IC) - Total instructions executed. Clock Frequency - Number of clock cycles per second. Clock Cycle Time - Time for one clock cycle. CPI - Average clock cycles per instruction. Execution Time - Total time to execute a program. MIPS - Million Instructions Per Second. IPC - Instructions Per Cycle. Speedup - Performance improvement over another processor.

Formulas

Metric	Formula
CPU Execution Time	IC × CPI × Clock Cycle Time
MIPS	Instruction Count / (Execution Time × 10^6)
IPC	1 / CPI
Speedup	Old Execution Time / New Execution Time

Algorithm

1. Read IC, CPI and Clock Frequency.
2. Calculate Clock Cycle Time = 1 / Clock Frequency.
3. Calculate CPU Execution Time.
4. Calculate MIPS.
5. Calculate IPC.
6. Compare processors.
7. Display results.

Complete C Program

```
#include<stdio.h>

int main()
{
    double IC,CPI,freq,cycleTime,execTime,MIPS,IPC;

    printf("Enter Instruction Count : ");
    scanf("%lf",&IC);
```

```
    printf("Enter CPI : ");
    scanf("%lf",&CPI);

    printf("Enter Clock Frequency (Hz): ");
    scanf("%lf",&freq);

    cycleTime = 1/freq;
    execTime = IC*CPI*cycleTime;
    MIPS = IC/(execTime*1000000);
    IPC = 1/CPI;

    printf("\nExecution Time = %lf seconds",execTime);
    printf("\nMIPS = %lf",MIPS);
    printf("\nIPC = %lf",IPC);

    return 0;
}
```

Sample Input

Instruction Count : 1000000
CPI : 2
Clock Frequency : 1000000000

Sample Output

Execution Time = 0.002000 seconds
MIPS = 500.000000
IPC = 0.500000

Worked Example

Given: IC = 1,000,000 instructions CPI = 2 Clock Frequency = 1 GHz Clock Cycle Time = 1 / 1,000,000,000 = 1 ns Execution Time = IC × CPI × Clock Cycle Time = 1,000,000 × 2 × 1 ns = 0.002 seconds IPC = 1 / 2 = 0.5 MIPS = 500

Time & Space Complexity

Time Complexity: O(1)
Space Complexity: O(1)

Applications

- Processor comparison
- Computer architecture research
- CPU performance analysis
- Benchmark testing

Advantages

- Easy performance evaluation.
- Helps compare processors.
- Useful in hardware design.

Viva Questions

What is CPI? - Clock Cycles Per Instruction.

What is IPC? - Instructions executed in one clock cycle.

What is MIPS? - Million Instructions Per Second.

What is benchmarking? - Measuring processor performance.

Why is execution time important? - It determines processor efficiency.